

INTERNSHIP PROJECT REPORT

YOLO11n-Based Space Debris and Satellite Detection Using Deep Learning

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By

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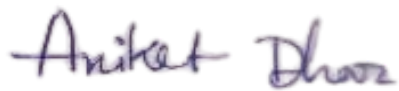
Bengal Institute Of Technology(2023-27)

BSERC Roll No. 13613

DECLARATION

We hereby declare that the project report entitled ” **YOLO11n-Based Space Debris and Satellite Detection Using Deep Learning**” submitted to the Bharat Space Education Research Centre(BSERC) for the Def-Space Summer Internship program is a record of original work done by us. The simulation models, expected outcome presented in this report have not been submitted elsewhere for any other internship or academic credit.

Dated: August 30, 2026

A handwritten signature in purple ink that reads "Aniket Dhar". The signature is written in a cursive, flowing style.

(Aniket Dhar)

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CONTENT

1. Introduction [5]

2.Literature Review 6]

3. Methodology [11]

4. Result [15]

5. Conclusion [21]

6. References [23]

1. Introduction

Earth's orbital environment contains a growing number of active satellites, inactive spacecraft, spent rocket stages, and fragments produced by previous missions. Even small debris particles can pose a serious collision risk because objects in orbit travel at very high relative velocities. Reliable detection and tracking of these objects is therefore important for maintaining space situational awareness.

Identifying objects in optical images is challenging because many targets occupy only a small number of pixels and may have weak contrast against dark or uneven backgrounds. Variations in illumination, image noise, and target size can also make conventional image-processing techniques difficult to apply consistently. Deep learning-based object detectors provide an alternative by learning relevant features from annotated training images rather than relying on manually designed image processing rules. YOLO-based detectors are particularly useful for this task because they predict object classes and bounding-box coordinates within a single inference stage.

In this study, a lightweight YOLO11n model was trained to detect two classes, satellites and space debris. The model was trained for 50 epochs using images resized to 640×640 pixels. Its performance was evaluated using precision, recall, F1-score, mAP@50. Detection results were also examined using predicted bounding boxes and confidence scores.

The class wise results provide a clearer view of the model's detection behavior, especially for small or faint objects. The findings can be used to identify limitations in the current dataset and model configuration and to determine possible improvements in small-object detection, training data, and subsequent real-time tracking.

2. Literature Review

2.1 Introduction

This chapter reviews the existing body of research relevant to space-debris and space-object detection. As the orbital environment becomes increasingly congested, tracking these objects has become a critical challenge for global space agencies. To provide context for the proposed methodology, this review covers the fundamentals of space debris detection and traces the evolution of computer vision from traditional image processing techniques to advanced deep learning architectures. Specifically, it explores the progression of Convolutional Neural Networks (CNNs) and focuses heavily on YOLO-based (You Only Look Once) object detection methods, establishing the theoretical foundation for using a lightweight YOLO11 model in this project.

2.2 Space Debris Detection

Space debris encompasses all defunct, human-made objects currently orbiting the Earth, ranging from dead satellites and spent rocket stages to tiny fragments created by collisions and degradation. Detecting and monitoring these objects is of paramount importance to ensure the safety of active space assets, including the International Space Station (ISS) and operational satellites. The exponential growth of orbital objects, driven by historical fragmentation events and the recent deployment of commercial satellite mega-constellations, has significantly increased the risk of the Kessler syndrome—a theoretical scenario where cascading collisions render certain orbits unusable.

Observation of these objects is generally categorized into ground-based and space-based systems. Ground-based observation primarily utilizes radar for low Earth orbit (LEO) and optical telescopes for higher orbits. Space-based observation involves using sensors mounted on orbiting satellites to track debris without atmospheric interference. Among these, optical and image-based detection is highly favored for its cost effectiveness and ability to capture the sunlight reflecting off orbital objects.

However, optical detection faces severe difficulties. Space debris objects are often extremely small, faint, and travel at high velocities. In captured images, they suffer from low contrast and are frequently obscured by severe sensor noise and complex backgrounds, such as dense star fields or the Earth's albedo. Traditional approaches for debris detection relied heavily on classical image processing techniques, such as frame differencing, background subtraction, thresholding, and morphological operations. While useful for simple scenarios, these traditional methods struggle to dynamically adapt to the complex, noisy, and low-contrast conditions typical of space imagery, thereby establishing the need for more robust, learning-based detection systems.

2.3 Deep Learning for Object Detection

The field of computer vision has undergone a massive paradigm shift. Early object detection relied on a progression starting from traditional feature-based methods (such as HOG or SIFT), which required manual feature engineering. These features were then fed into traditional Machine Learning classifiers like Support Vector Machines (SVMs). The breakthrough came with the advent of Convolutional Neural Networks (CNNs), which automated the feature extraction process by learning hierarchical representations directly from pixel data. This evolution ultimately led to modern Deep Learning frameworks dedicated to Object Detection.

Unlike simple image classification, object detection performs three distinct but interconnected tasks simultaneously. First, it identifies the object within an image, distinguishing the target from the background. Second, it determines its class, categorizing what the object actually is (e.g., satellite, debris). Finally, it localizes it using a bounding box, outputting the exact spatial coordinates (x, y, width, height) of the target within the frame. This end-to-end learning capability makes deep learning exceptionally suited for isolating faint objects in complex space imagery.

2.4 YOLO-Based Object Detection

Among modern object detection frameworks, the YOLO (You Only Look Once) architecture has emerged as a state-of-the-art solution. Introduced by Redmon et al. (2016), YOLO revolutionized the field by framing object detection as a single regression problem rather than a classification task on regional proposals.

As a one-stage detection approach, a single neural network passes over the entire image once, dividing it into a grid. Each grid cell is responsible for predicting bounding boxes and class probabilities simultaneously. This approach allows the network to reason globally about the full image and all the objects in it. The primary advantages of YOLO are its incredibly fast inference speeds, allowing for real-time detection, and its strong generalized representations of objects.

Since its original inception, the YOLO architecture has seen continuous evolution, with successive versions introducing feature pyramid networks (FPNs), improved loss functions, and anchor-free detection mechanisms. These improvements have vastly enhanced YOLO's application to small-object detection, a critical requirement for spotting space debris.

The model utilized in this project is YOLO11, one of the latest iterations in the YOLO family (Jocher & Qiu, 2024). YOLO11 builds upon the speed and accuracy of its predecessors while optimizing computational efficiency, offering varied model sizes, including lightweight (nano) versions that are highly effective for deployment in resource-constrained environments.

2.5 Review of Previous Research

Author & Year	Objective	Method	Key Finding	Limitation
Yuan et al. (2023)	Detect small/weak space objects	CS-YOLOv5	Improved detection of small and weak objects	Small-object detection remains challenging
Zhang et al. (2024)	Detect spaceborne dynamic objects	YOLO-Dynamic	Improved detection performance over baseline YOLOv8s	Focused on specific space-object conditions
Guo et al. (2025)	Space-debris detection	Enhanced YOLOv8	Improved cross-scale feature detection	Debris remains difficult in complex backgrounds
Improved YOLO11 (2025)	Space-object detection	YOLO11-based approach	Demonstrated suitability of YOLO11 for space objects	Requires further evaluation across datasets
Kerr & Arandjelović (2026)	On-board space-debris detection	Lightweight/compressed YOLO approach	Investigated efficient deployment	Different model/dataset from the present study

2.6 Comparative Analysis

Approach	Advantage	Limitation
Traditional image processing	Simple, computationally inexpensive	Requires handcrafted features
CNN classification	Good feature learning	Does not directly provide object localization
Two-stage detectors	High detection capability	Generally more computationally expensive
YOLO	Fast detection + localization	Performance can vary for small objects
Lightweight YOLO	Lower computational requirements	May sacrifice some detection performance

2.7 Research Gap

While the literature demonstrates significant progress in space object detection, several critical gaps remain:

- **Size and Visibility:** Space debris can be extremely small and notoriously difficult to detect, especially given that optical images often suffer from low contrast, severe sensor noise, and complex star-field backgrounds.
- **Computational Complexity:** Many of the highly accurate deep learning models (like two-stage detectors) are computationally complex and heavy, making them difficult to deploy on resource-constrained satellite hardware or field equipment.
- **Narrow Focus of Existing Studies:** Several existing studies focus exclusively on video tracking using multi-frame algorithms, or propose highly customized, modified architectures that are difficult to reproduce and deploy efficiently.

Consequently, there is a clear scope for evaluating a state-of-the-art, off-the-shelf lightweight model, specifically the YOLO11n model, for satellite and debris detection using static, image-based object detection. This project addresses this gap by training and evaluating the YOLO11n model to determine if a highly lightweight, single-stage detector can successfully overcome the noise and low-contrast challenges of space imagery while maintaining the speed and low computational footprint required for modern aerospace applications.

2.8 Chapter Summary

This chapter reviewed the critical need for space debris detection and the evolution of the technologies used to achieve it. Traditional optical observation methods, while foundational, are insufficient for the noisy and complex nature of space imagery. The literature shows a clear progression toward deep learning, where YOLO-based one-stage detectors offer the best balance of speed and accuracy. However, a gap remains regarding the application of the newest lightweight architectures to static space imagery. Building upon these insights, the next chapter details the methodology utilized in this research, outlining the dataset preparation, model configuration, and training pipeline for implementing YOLO11n to detect space objects.

3. Methodology

3.1 Dataset Collection

A dataset containing images of **satellites and space debris** was used for developing the object-detection model. The images represent objects captured against space backgrounds and were prepared for training a deep-learning-based computer vision model. The dataset was organized into training, validation, and testing subsets so that the model could be trained and subsequently evaluated on previously unseen images.

Dataset Source: The dataset used in this project was obtained from Space Debris Computer Vision Dataset by [Capstone Design](#). The original dataset can be accessed through the following link:

[\[https://universe.roboflow.com/capstone-design-ecfaa/space-debris-pidgc\]](https://universe.roboflow.com/capstone-design-ecfaa/space-debris-pidgc).

3.2 Data Preprocessing and Annotation

The collected images were inspected and prepared before model training. Each target object in the images was annotated using **bounding boxes**, with the annotations identifying the location and class of the object.

The dataset was converted into the format required by **YOLO (You Only Look Once)**. Each annotation contains the object class and the normalized bounding-box coordinates:

- Object class
- Bounding-box center xx
- Bounding-box center yy
- Bounding-box width
- Bounding-box height

The images and corresponding labels were then organized into the required directory structure for YOLO training.

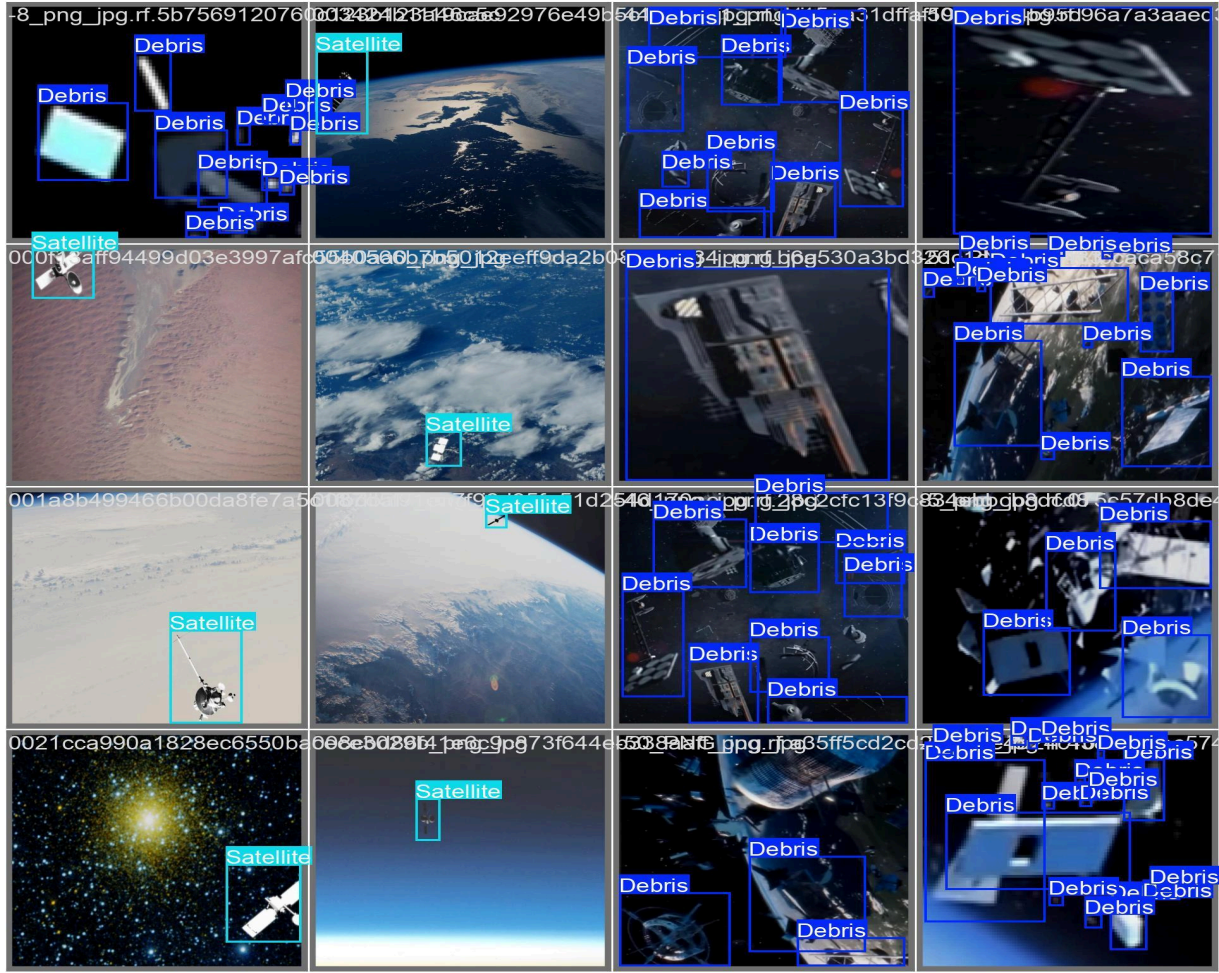


Figure 3.2: Ground-truth annotations for a validation image batch

3.3. Dataset Splitting

The prepared dataset was divided into three subsets:

- **Training set** – used to train the model and learn visual features of satellites and debris.
- **Validation set** – used during training to monitor model performance and help assess generalization.
- **Test set** – used after training to evaluate the final model on unseen data.

This separation prevents the evaluation from being based on images that the model has already used for learning.

3.4 YOLO Model Selection

A YOLO-based object-detection architecture was selected because the objective of the project is not simply to classify an image but to locate and identify objects within an image using bounding boxes.

The YOLO framework performs object detection by processing an image and predicting the object class and bounding-box coordinates in a single detection pipeline. This makes it suitable for applications where both detection accuracy and computational efficiency are important.

3.5 Model Training

The prepared dataset was used to train the YOLO model. During training, the model learned the visual characteristics and spatial features associated with the target objects.

The model was trained for **50 epochs**. During each epoch, the training images were passed through the network, predictions were compared with the ground-truth annotations, and the model weights were updated to reduce the detection loss.

The major losses monitored during training included:

- **Box loss** – measures the error in predicted bounding-box localization.
- **Classification loss** – measures errors in identifying the object class.
- **Distribution Focal Loss (DFL)** – contributes to accurate localization of bounding-box coordinates.

3.6 Model Validation

Validation was performed during the training process using the separate validation dataset. This allowed the model's detection performance to be monitored on images that were not directly used to update the model weights.

Performance was assessed using standard object-detection metrics such as:

- **Precision**
- **Recall**
- **mAP@50**
- **mAP@50–95**
- **F1-score**

3.7 Model Testing

After training was completed, the final trained model was evaluated using the **test dataset**. The test images were not used during model training, allowing the final model's ability to generalize to the unseen images to be assessed.

The model generated predictions consisting of:

1. Detected object class
2. Bounding-box coordinates
3. Confidence score

3.8 Performance Evaluation

The final model was evaluated using the generated performance plots and detection metrics. In particular, the **Precision Recall curve, F1 curve, confusion matrix, and mAP results** were examined.

The evaluation focused on determining how accurately the model could:

- Detects satellites/space debris.
- Correctly classify detected objects.
- Localize objects using bounding boxes.
- Minimize false detections.
- Detect objects that were actually present.

The **mAP@50–95** metric was also considered because it provides a stricter evaluation across multiple Intersections over Union (IoU) thresholds.

3.9 Detection Visualization

Finally, the trained model was applied to test images to visually verify its predictions. Bounding boxes were drawn around detected objects along with their predicted class and confidence score.

This visual analysis provided an additional qualitative assessment of the model and helped identify cases such as missed detections, incorrect detections, and inaccurate bounding box localization.

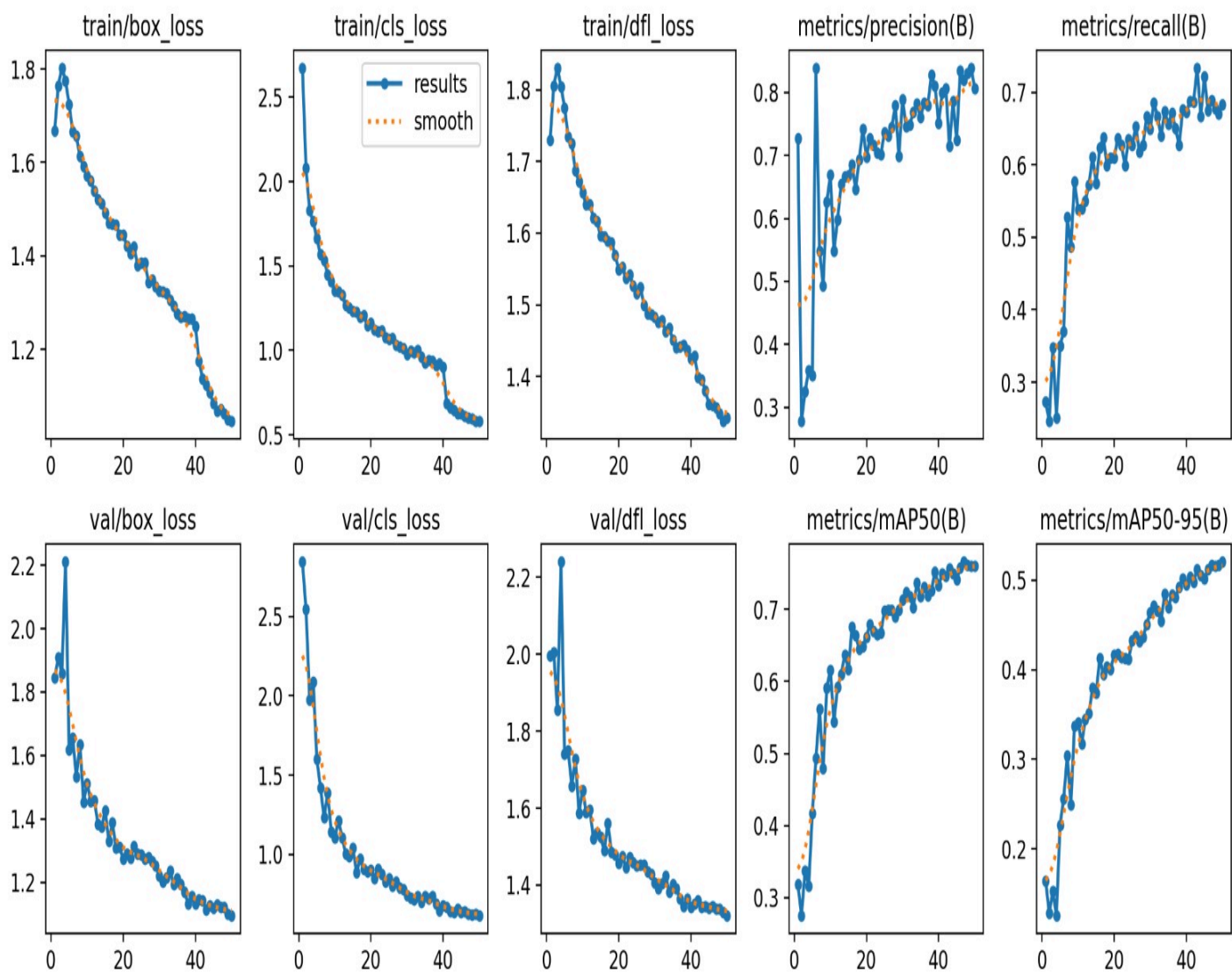
4.Results

4.1 Model Training Results

The proposed satellite and space-debris detection model was trained using **YOLO11n** for **50 epochs** with an input image size of **640×640 pixels**. The model was trained on a dataset containing **3,727 training images**, while the validation set contained **702 images and 830 object instances**. No corrupt images were reported in either the training or validation datasets.

During training, the model progressively learned to identify the two target classes, **Debris** and **Satellite**. The training process monitored three major loss components: **box loss, classification loss, and Distribution Focal Loss (DFL)**. By the final epoch, the recorded losses were approximately **1.046 for box loss, 0.583 for classification loss, and 1.343 for DFL**.

The complete training process took approximately **0.915 hours**, after which the trained model weights were saved for subsequent evaluation.

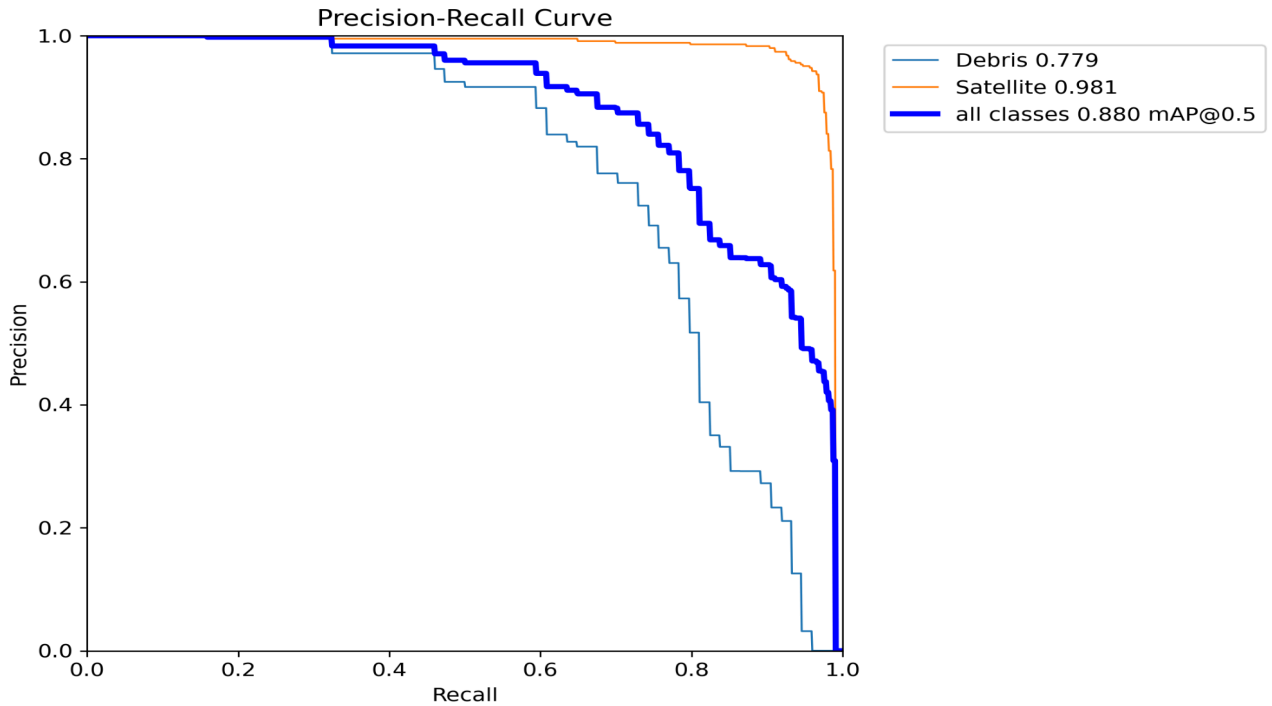


4.2 Overall Detection Performance

The final validation of the trained **YOLO11n** model produced the following overall results:

Metric	Result
Precision	0.808
Recall	0.683
mAP@50	0.760
mAP@50–95	0.520

These results indicate that the model achieved a **precision of 80.8%**, meaning that a high proportion of its positive detections were correct. The **recall of 68.3%** indicates that the model successfully detected a substantial proportion of the objects present in the validation dataset. The model achieved **76.0% mAP@50** and **52.0% mAP@50–95**, demonstrating reasonable object-detection performance, with the stricter mAP@50–95 metric indicating that bounding-box localization can still be improved.



4.3 Class-wise Performance

The model performed differently for the two target classes.

Class	Precision	Recall	mAP@50	mAP@50–95
Debris	0.659	0.432	0.544	0.323
Satellite	0.957	0.934	0.976	0.718

The **Satellite** class achieved substantially stronger performance, with **95.7% precision**, **93.4% recall**, **97.6% mAP@50**, and **71.8% mAP@50–95**. This indicates that the model was highly effective at detecting and localizing satellites in the validation images.

In comparison, the **Debris** class achieved **65.9% precision**, **43.2% recall**, **54.4% mAP@50**, and **32.3% mAP@50–95**. The lower performance suggests that debris detection was considerably more challenging than satellite detection. This may be

associated with the smaller appearance, low contrast, irregular shapes, and visual similarity of debris objects in the images.

4.4 F1-Score Analysis

The F1-confidence analysis was used to identify a confidence threshold that provides a balance between precision and recall. The F1-confidence curve showed a **maximum overall F1-score of approximately 0.84 at a confidence threshold of 0.469**.

The class-wise curves also showed that the model performed considerably better for **Satellite** than for **Debris**. The Satellite curve reached a substantially higher F1-score, whereas the Debris curve remained lower across the confidence range.

This result is consistent with the class-wise mAP results: satellite detection is the stronger component of the model, while debris detection remains the main area requiring improvement.

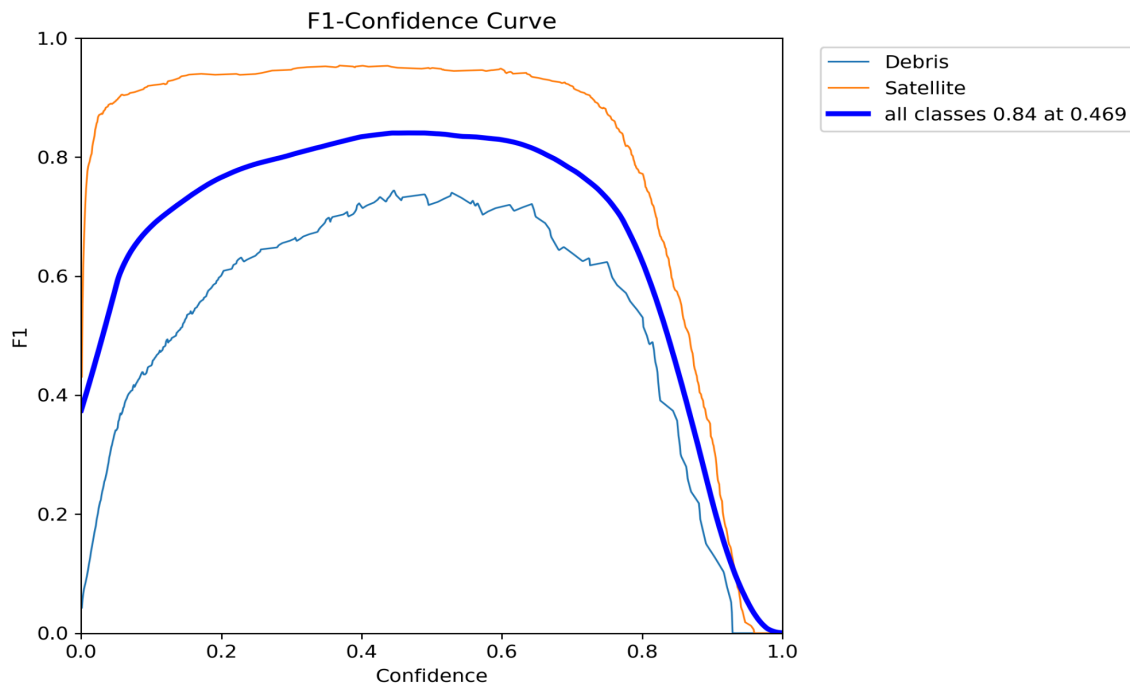


Figure 4.4: *F1_curve*

4.5 Confusion Matrix Analysis

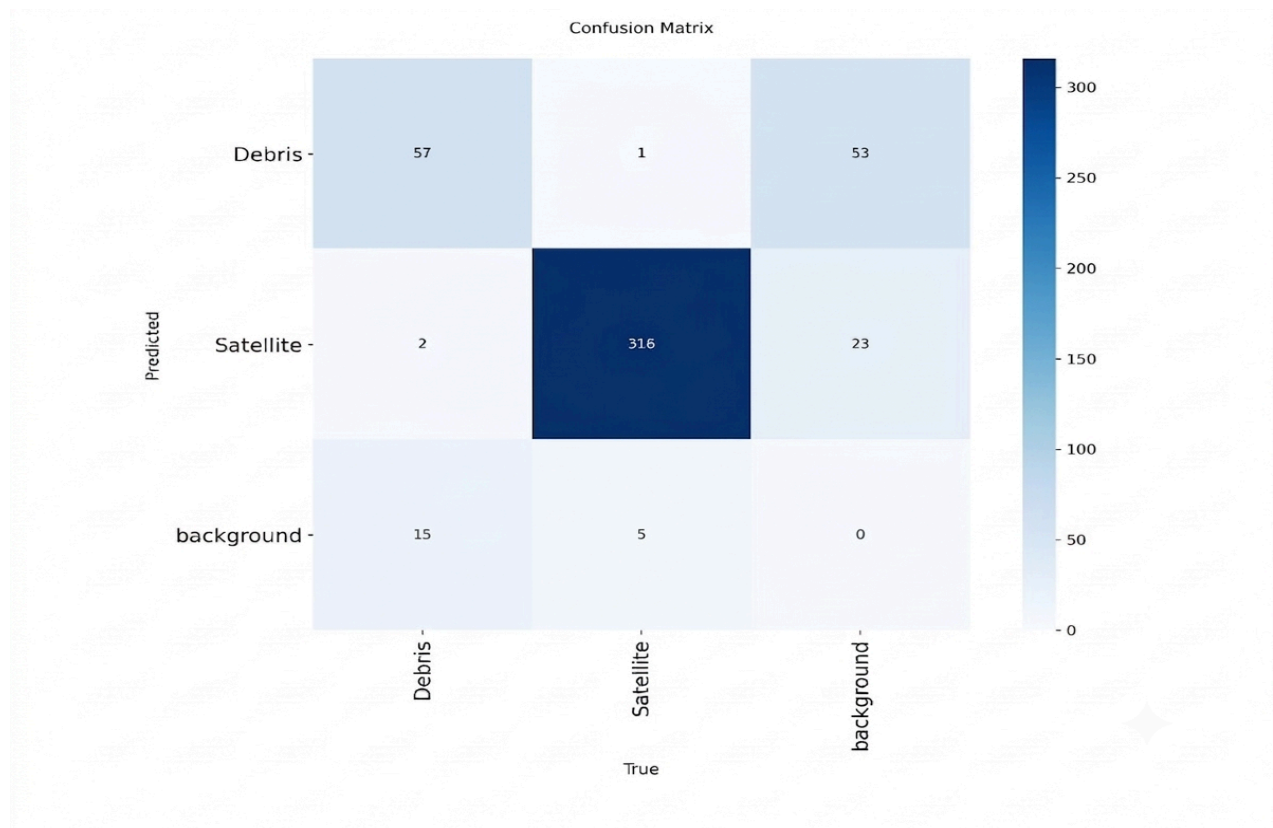


Figure 4.5: *Confusion Matrix of the YOLO11n Model*

The confusion matrix was used to analyse the class-wise prediction behaviour of the YOLO11n model for Debris and Satellite. The matrix compares the true object classes with the predicted classes and also shows predictions associated with the background.

The model correctly predicted 57 debris instances and 316 satellite instances. The matrix also shows some misclassification and missed detections, particularly for the Debris class. In comparison, the Satellite class shows substantially stronger prediction performance, which is consistent with the higher precision, recall, and mAP values obtained for satellites.

These results further indicate that Debris detection is more challenging than Satellite detection.

4.5 Detection Visualization

The trained model was also applied to satellite images to visually verify its object-detection capability. The model produced bounding boxes around detected objects along with their predicted class labels and confidence scores.

The detection results demonstrate that the model can identify satellites across different image backgrounds, orientations and scales. The bounding-box predictions provide both the **location and class of the detected object**, making the system suitable for an object-detection-based satellite and space-debris monitoring application.

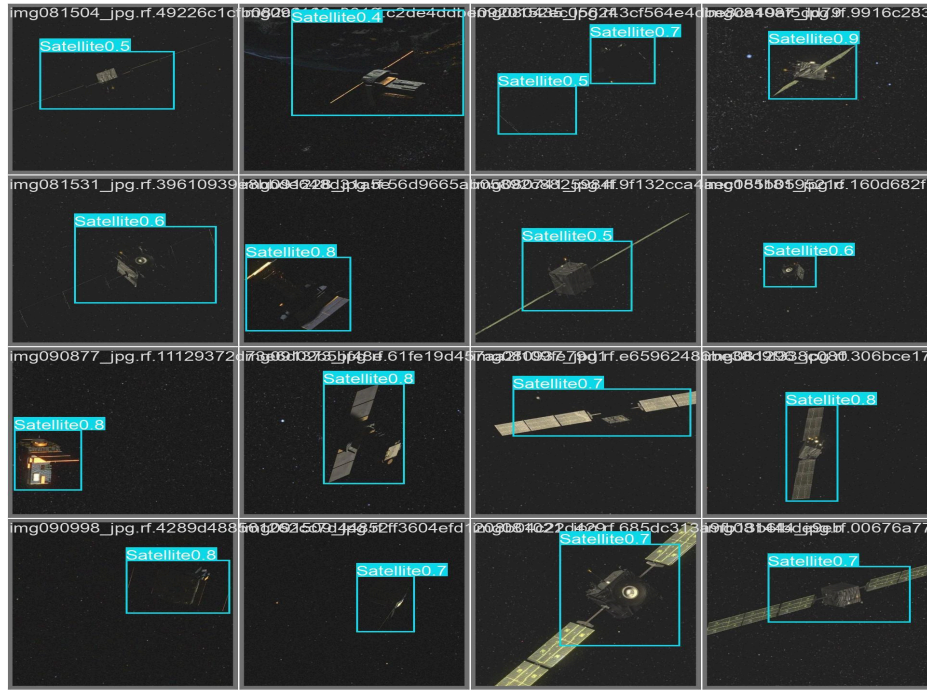


Figure 4.6: *Validation batch predictions generated by the YOLO11n model*

4.6 Discussion of Results

Overall, the experimental results demonstrate that the YOLO11n model can successfully detect and localize **satellites and space debris**, but its performance is not uniform across the two classes.

The strongest result was obtained for the **Satellite** class, which achieved a **97.6% mAP@50** and **93.4% recall**. In contrast, the **Debris** class achieved only **54.4% mAP@50** and **43.2% recall**. Therefore, the overall performance is strongly influenced by the difficulty of detecting debris.

The overall **mAP@50 of 76.0%** demonstrates that the model provides useful detection capability, while the lower **mAP@50–95 of 52.0%** indicates that more precise localization remains a challenge. Future improvements should therefore focus particularly on **debris detection, dataset balancing, higher-quality annotations, additional training data, and model optimization**.

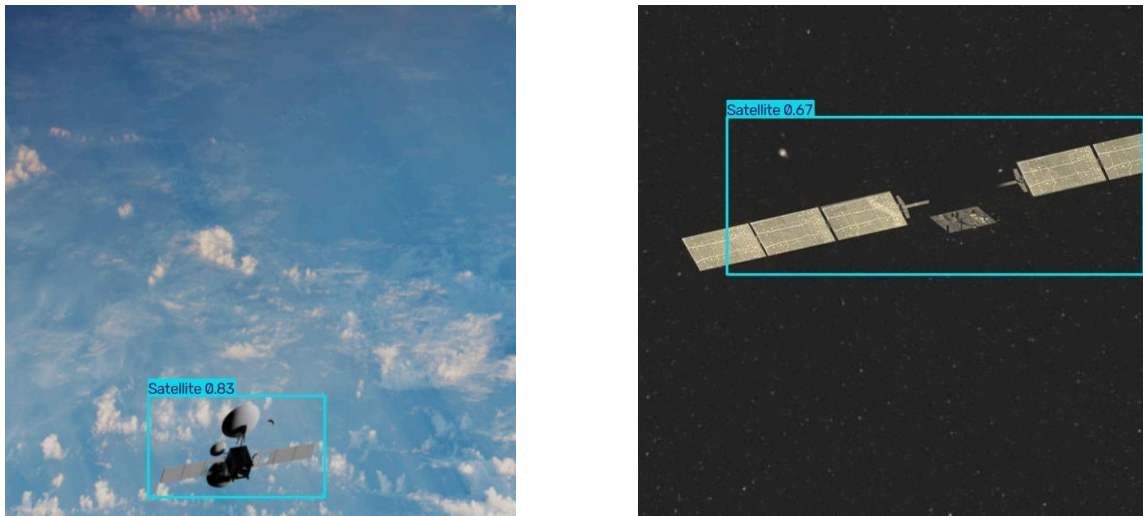


Figure 4.6: *Sample Detection Results of YOLO11n*

❖ **Key result to highlight**

YOLO11n achieved 80.8% precision, 68.3% recall, 76.0% mAP@50, and 52.0% mAP@50–95 after 50 epochs. Satellite detection was substantially stronger than debris detection.

5. Conclusion

The project successfully developed and evaluated a **YOLO11n-based object detection model** for the detection and localization of **satellites and space debris**. The model was trained for **50 epochs** using annotated images and was evaluated using standard object-detection metrics.

The trained model achieved an overall **precision of 80.8%, recall of 68.3%, mAP@50 of 76.0%, and mAP@50–95 of 52.0%**. The results demonstrate that the model is capable of detecting and localizing target objects using bounding boxes.

Class-wise evaluation showed that **satellite detection was considerably stronger than debris detection**, with satellites achieving **97.6% mAP@50**, compared with **54.4% for debris**. The difference indicates that detecting small, irregular, or visually less distinct debris remains the primary challenge of the current system.

Overall, the study demonstrates the **feasibility of using YOLO-based deep learning for satellite and space-debris detection**. However, further improvements in dataset size and diversity, debris representation, annotation quality, model optimization, and training strategies could improve detection and localization performance.

Future Scope

- Increase the size and diversity of the dataset.
- Include more challenging and small debris objects.
- Improve annotation quality and class balance.
- Experiment with different YOLO architectures and hyperparameters.
- Apply data augmentation to improve model generalization.
- Evaluate the model on a larger and more diverse independent test dataset.
- Explore real-time or automated space-object monitoring applications

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